ADUMBRATIO DEMONSTRATIONIS THEOREMATIS ARITHMETICI MAXIME UNIVERSALIS.

AUCTORE

C. F. DEGEN.

Conventui exhibuit die 7. Octobris 1818.

§. 1. Jamdudum nota sunt Analyscos diophanteae cultoribus sequentia theoremata:

I. $(p^2 + q^2)\,(r^2 + s^2) = (pr \pm qs)^2 + (ps \mp qr)^2$,

II. $(fp^2 + gq^2)(fr^2 + gs^2) = (fpr \pm gqs)^2 + fg\,(ps \mp qr)^2$,

unde, posito $f = 1$ et $g = a$, sequitur

III. $(p^2 + aq^2)\,(r^2 + as^2) = (pr \pm aqs)^2 + a\,(ps \mp qr)^2$,

nec non haec:

IV. $(P^2 + Q^2 + R^2 + S^2)\,(p^2 + q^2 + r^2 + s^2)$
$= (Pp + Qq + Rr + Ss)^2 + (Pq - Qp + Rs - Sr)^2$
$+ (Pr - Qs - Rp + Sq)^2 + (Ps + Qr - Rq - Sp)^2$,

quod summo *Eulero* acceptum refertur, et generalius hoc:

V. $(P^2 - BQ^2 - CR^2 + BCR^2)\,(p^2 - Bq^2 - Cr^2 + BCs^2)$
$= (Pp + BQq \pm CRr \pm BCSs)^2 - B(Pq + Qp \pm CRs \pm CSr)^2$
$- C(Pr - BQs \pm Rp + BSq)^2 + BC\,(Ps - Qr \pm Rq \mp Sp)^2$,

quod ill. *Lagrange* (V. Mémoires de Berlin, année 1770. p. 133 in fine), praeclaris suis inventis addidit, quodque omnia praecedentia in se complectitur. Idem vero etiam alio respectu est notatu dignum, cum ita enunciari possit:

Si numerus formae $P^2 - BQ^2 - CR^2 + BCS^2$ *per alium numerum ejusdem formae, puta* $p^2 - bq^2 - cr^2 + bcs^2$ multiplicetur, prodibit factum ejusdem formae, scil. $F = V^2 - BX^2 - CY^2 + BCZ^2$, sumendo

Digitized by Google

$$\begin{aligned} V &= Pp + BQq \pm CRr \pm BCSs \\ X &= Pq + Qp \pm CRs \pm CSr \\ Y &= Pr - BQs \pm Rp \mp BSq \\ Z &= Ps - Qr \pm Rq \mp Sp. \end{aligned}$$

Specialiori *summae* generaliorem *formae* notionem, quadratis *simplicibus* quadratorum *multipla* substituit summus ille Analysta. Methodum vero inveniendi et demonstrandi lectori curioso reliquit. Alio vero loco, scil. in Additamentis ad *Euleri* Algebram (ad Cap. 11^mum et 12^mum Anal. indeterm.) celeberrimus idem Autor alias functiones algebraicas contemplatus est, quarum producta ipsis similia sunt, quales v. c. $x^2 + axy + by^2$, $x^3 + ax^2y + bxy^2 + cy^3$ etc. quam speculationem equidem ad formas mixtas sc. non homogeneas, quales est v. c. $x^2 + axy + by^2 + ex + dy + e$ extendi et ad solutionem aequationum indeterminatarum secundi ordinis cum successu applicari posse observavi, cujus rei specimen in alia dissertatiuncula exhibere conatus sum.

§. 2. Prout autem in adscensu Matheseos ad maxime sublimia haud raro fit ut, quae maxime nobis videantur generalia, universaliorum non esse nisi consectaria in posterum deprehendamus, ita et hic probabile fiet, theorema *Lagrangianum* non esse nisi anticam catenae infinitae, aliquot annulos continentem, partem, atque monstrare conabor, *esse functiones hujus formae:*

$$\begin{aligned} &p^2 + aq^2 + br^2 + abs^2 + ct^2 + acu^2 + bcv^2 + abcx^2 + dy^2 \\ &+ adz^2 + bdy'^2 + cdq'^2 + abdr'^2 + acds'^2 + bcdt'^2 + abcdu'^2 \\ &+ ev'^2 + aesx'^2 + beyf'^2 + cez'^2 + \text{etc. etc.} \end{aligned}$$

hac proprietate insigni praeditas, ut formata ex hujus generis functionibus producta ipsis factoribus sint similia; id quod vel per transennam conspexisse admodum est volupe; quare in re tam ardua jam plurimum praestitisse mihi videor, si theorema illud ad terminum $a\,b\,c\,x^2$ usque extenderim et analogiae, cujus veritas ex ipsa inventionis methodo satis elucet, quod reliquum est, superstruxerim.

Digitized by Google

§. 3. Ante omnia necesse est extensionem theorematis *Euleriani* ad summam octo quadratorum ostendere: Fit nempe

$$(P^2+Q^2+R^2+S^2+T^2+U^2+V^2+X^2)(p^2+q^2+r^2+s^2+t^2+u^2+v^2+x^2)=\Pi$$
$$=(Pp+Qq+Rr+Ss+Tt+Uu+Vv+Xx)^2$$
$$+(Pq-Qp+Rs-Sr+Tu-Ut+Vx-Xv)^2$$
$$+(Pr-Qs-Rp+Sq\mp Tv\pm Ux\pm Vt\mp Xu)^2$$
$$+(Ps+Qr-Rq-Sp\pm Tx\pm Uv\mp Vu\mp Xt)^2$$
$$+(Pt-Qu\pm Rv\mp Sx-Tp+Uq\mp Vr\pm Xs)^2$$
$$+(Pu+Qt\mp Rx\mp Sv-Tq-Up\pm Vs+Xr)^2$$
$$+(Pv-Qx\pm Rt\pm Su\pm Tr\mp Us-Vp+Xq)^2$$
$$+(Px+Qv\pm Ru\pm St\mp Ts\mp Ur-Vq-Xp)^2,$$

cujus aequalitatis dispositionem ita me instituisse spero, ut etiam minus sagacibus de veritate ejus facile constare possit. Sagaciores eandem facile ad sedecim etc. quadratorum summam extensuros esse mihi persuasum est. Prolixa igitur aeque ac taediosa descriptione supersedeo. Circa ipsum hoc theorema vero notasse oportet, deceptum iri qui hujus et *Fermatiani* illius, quo statuitur, *omnem numerum esse quatuor pauciorumve quadratorum summam*, aliquam ut aiunt *identitatem* ideo statuerent, quod sumto pro lubitu $\Pi=\Sigma+\Theta$, secundum *Fermatium* Σ in $\alpha^2+\beta^2+\gamma^2+\delta^2$, Θ in $\varepsilon^2+\zeta^2+\eta^2+\theta^2$ resolvi possit, adeoque semper sit Π octo quadratorum summa. Exclusis autem (e quadratorum realium classe) cyphris, non semper habebuntur octena illa quadrata, quae nostra formula immediate ita exhibet, ut nec conjecturae nec vagis tentaminibus locus relinquatur; idem de theoremate *Euleriano* valere protinus perspicitur. Praeterea vero aliud est *resolutionis possibilitatem enunciare*, aliud *ipsam resolutionem commonstrare*.

§. 4. His praemissis me ad praecedentis theorematis amplificationem accingo, indeterminatis quos ajunt coëfficientibus in auxilium adcitis. Sic quod absque demonstratione proponere licuisset,

Digitized by Google

problematis instar veri nominis solutione munivisse praetuli. cum exinde aditus ad ulteriora pateat, quamplurimum laudi et utilitati certissimae adeoque peregregiae scientiae officere arbitratus eos, qui, cum non sint nisi secundi ordinis Geometrae, principum more se gerere gestiant, cum ne ipsos quidem hosce deceat nimia caligine tectos incedere (*).

Introducamus pro unitatibus factores $a, b, c \ldots g$; $\alpha, \beta, \gamma, \ldots \eta$; $\alpha', \beta', \gamma' \ldots \eta'$; $\alpha'', \beta'', \gamma'' \ldots \eta''$; ..... $\alpha^{VI}, \beta^{VI}, \gamma^{VI}, \ldots \eta^{VI}$ atque ponamus

$(P^2 + aQ^2 + bR^2 + cS^2 + dT^2 + eU^2 + fV^2 + gX^2)$ mult. per

$(p^2 + aq^2 + br^2 + cs^2 + dt^2 + eu^2 + fv^2 + gx^2)$ esse

$= \varrho^2 + a\sigma^2 + b\tau^2 + c\upsilon^2 + d\Phi^2 + e\chi^2 + f\psi^2 + g\omega^2$

$= (Pp + aQq + bRr + cSs + dTt + eUu + fVv + gXx)^2$

$+ a(Pq - \alpha Qp + \beta Rs - \gamma Sr + \delta Tu - \varepsilon Ut + \zeta Vx - \eta Xv)^2$

$+ b(Pr - \alpha' Qs - \beta' Rp + \gamma' Sq \mp \delta' Tv \pm \varepsilon' Ux \pm \zeta' Vt \mp \eta' Xu)^2$

$+ c(Ps + \alpha'' Qr - \beta'' Rq - \gamma'' Sp \pm \delta'' Tx \pm \varepsilon'' Uv \mp \zeta'' Vu \mp \eta'' Xt)^2$

$+ d(Pt - \alpha''' Qu + \beta''' Rv \mp \gamma''' Sx - \delta''' Tp + \varepsilon''' Uq \mp \zeta''' Vr \pm \eta''' Xs)^2$

$+ e(Pu + \alpha^{IV} Qt \mp \beta^{IV} Rx \mp \gamma^{IV} Sv - \delta^{IV} Tq - \varepsilon^{IV} Up \pm \zeta^{IV} Vs \pm \eta^{IV} Xr)^2$

$+ f(Pv - \alpha^{V} Qx \mp \beta^{V} Rt \pm \gamma^{V} Su \pm \delta^{V} Tr \mp \varepsilon^{V} Us - \zeta^{V} Vp + \eta^{V} Xq)^2$

$+ g(Px + \alpha^{VI} Qv \pm \beta^{VI} Ru \pm \gamma^{VI} St \mp \delta^{VI} Ts \mp \varepsilon^{VI} Ur - \zeta^{VI} Vq - \eta^{VI} Xp)^2$.

Quod autem coëfficientes α, β', γ'', δ''', ε^{IV}, ζ^{V} et η^{VI}, secundum diagonalem $Pp \ldots Xp$ dispositi et signo negativo affecti sint unitati aequales, vel levissima attentione perspicitur. Deinde cum productum illud, numerorum indeterminatorum $P, Q \ldots p, q \ldots$

(*) Nostra certe Urania ab illa *Virgilii* longe differt, quae (v. Aeneid. I. 411 — 414.)

. . . . *obscuro gradientes aëre saepsit,*
Et multo nebulae circum dea fudit amictu:
Cernere ne quis eos neu quis contingere posset,
Molirive moram aut veniendi poscere caussas.

Apertos atque simplices, non vanam ex admiratione vulgi laudem captantes, honestos et strenuos veritatum suarum cultores templa sua intrare jubet. Antiquae huic legi morem gerere, quid sanius, quid pulchrius?

Digitized by Google

quadrata tantum contineat, patet coëfficientes productorum $QRqr$, $QSqs$, ... $RSrs$, $RTrt$, ... $STst$, $SVsv$ ... etc. non-quadratorum nihilo esse aequales, unde sequentes 21 aequationes nanciscimur:

Scil. e coëfficientibus producti

1) $QRqr$ aequationem $ab = c\alpha''\beta''$
2) $QSqs$ . . . $ac = b\alpha'\gamma'$
3) $QTqt$ . . . $ad = e\alpha^{IV}\delta^{IV}$
4) $QUqu$ . . . $ae = d\alpha'''\varepsilon'''$
5) $QVqv$ . . . $af = g\alpha^{VI}\zeta^{VI}$
6) $Qxqx$ . . . $ag = f\alpha^{V}\eta^{V}$

7) $RSrs \ldots bc = a\beta\gamma$
8) $RTrt \ldots bd = f\beta^{V}\delta^{V}$
9) $RUru \ldots be = g\beta^{IV}\varepsilon^{IV}$
10) $RVrv \ldots bf = d\beta''\zeta'''$
11) $RXrx \ldots bg = e\beta^{IV}\eta^{IV}$

12) $STst$ . . . $cd = g\gamma^{VI}\delta^{VI}$
13) $SUsu$ . . . $ce = f\gamma^{V}\varepsilon^{V}$
14) $SVsv$ . . . $cf = e\gamma^{IV}\zeta^{IV}$
15) $SXsx$ . . . $cg = d\gamma'''\eta'''$

16) $TUtu \ldots de = a\delta\varepsilon$
17) $TVtv \ldots df = b\delta'\zeta'$
18) $TXtx \ldots dg = c\delta''\eta''$

19) $UVuv$ . . . $ef = c\varepsilon''\zeta''$
20) $UXux$ . . . $eg = b\varepsilon'\eta'$.

21) $VXvx \ldots fg = a\zeta\eta$

§. 5. Est autem numerus coëfficientium indeterminatorum $\alpha, \beta, \gamma, \ldots = 49$; quare, ob 21 aequationes illas et 7 priores e membrorum diagonalium indole derivatas, 21 horum coëfficientium nostro plane arbitrio reliquuntur, qua libertate ita utemur, ut formas fractas evitemus. Statuamus hunc in finem

C) $\alpha'' = \beta'' = \alpha^{IV} = \delta^{IV} = \alpha^{VI} = \zeta^{VI} = 1$, et erit

22) $c = ab$; 23) $e = ad$; 24) $g = af$ (ex aequatt. 1, 3, 5).

Quare multiplicando aequat. 1^{mam} per 2^{dam}, 3^{tiam} per 4^{tam}, 5^{tam} per 6^{tam}, adipiscimur

25) $a^2 = \alpha'\gamma'$; 26) $a^2 = \alpha'''\varepsilon'''$; 27) $a^2 = \alpha^{V}\eta^{V}$ unde colligo

28) $\alpha' = \gamma' = \alpha''' = \varepsilon''' = \alpha^{V} = \eta^{V} = a$.

29) $cb = ab^2 = a\beta\gamma$ (7, 22), sive $b^2 = \beta\gamma$; ergo 30) $\beta = \gamma = b$

31) $de = ad^2 = a\delta\varepsilon$ (16, 23), sive $d^2 = \delta\varepsilon$; ergo 32) $\delta = \varepsilon = d$

33) Ponendo porro $\beta^{V} = \delta^{V} = 1$, ex aequatione 8va emergit $f = bd$. Hinc ob

Digitized by Google

34) $bf = b^2 d = d\beta''' \zeta'''$ sc. $b^2 = \beta''' \zeta'''$. fit $\beta''' = \zeta''' = b$
(Ex aequat. 10, 33.)

35) $be = g\beta^{VI}\varepsilon^{VI}$ sc. $bad = af\beta^{VI}\varepsilon^{VI} = abd\beta^{VI}\varepsilon^{VI}$ (9, 23, 24, 33); ergo $\beta^{VI} = \varepsilon^{VI} = 1$.

36) $bf = d\beta^{IV}\eta^{IV}$ sc. $b^2 = \beta^{IV}\eta^{IV}$; ergo $\beta^{IV} = \eta^{IV} = b$ (11, 24, 23).

37) $cd = g\gamma^{VI}\delta^{VI}$ sc. $bd = f\gamma^{VI}\delta^{VI}$; ergo $\gamma^{VI} = \delta^{VI} = 1$.
(12, 22, 24, 33)

38) $ce = f\gamma^{V}\varepsilon^{V}$ sc. $a^2 bd = f\gamma^{V}\varepsilon^{V}$; ergo $\gamma^{V} = \varepsilon^{V} = a$.
(13, 22, 23, 33)

39) $cf = e\gamma^{IV}\zeta^{IV}$, sc. $bf = b^2 d = d\gamma^{IV}\zeta^{IV}$; inde $\gamma^{IV} = \zeta^{IV} = b$
(14, 22, 23, 33)

40) $cg = d\gamma'''\eta'''$ sc. $a^2 bf = a^2 b^2 d = d\gamma'''\eta'''$;

inde $\gamma''' = \left\{\begin{smallmatrix} \gamma^2 \\ \beta \\ \alpha^2 \end{smallmatrix}\right\}$, $\eta''' = \left\{\begin{smallmatrix} b^2 \\ \gamma b \\ a^2 \end{smallmatrix}\right\}$ sc. $\gamma''' = \left\{\begin{smallmatrix} a \\ ab^2 \\ b \\ ba^2 \end{smallmatrix}\right\}$ et $\eta''' = \left\{\begin{smallmatrix} ab^2 \\ a \\ a^2 b \\ b \end{smallmatrix}\right\}$ (15)

41) $df = b\delta'\zeta'$ sc. $bd^2 = b\delta'\zeta'$; ergo $\delta' = \zeta' = d$. (17)

42) $dg = c\delta''\eta''$ sc. $df = b\delta''\eta''$ sc. $bd^2 = b\delta''\eta''$; ergo $\delta'' = \eta'' = d$ (18)

43) $ef = c\varepsilon''\zeta''$ sc. $df = bd^2 = \varepsilon''\zeta''$; hinc $\varepsilon'' = \zeta'' = d$ (19)

44) $eg = b\varepsilon'\eta'$ sc. $a^2 df = a^2 bd^2 = b\varepsilon'\eta'$, ergo $\varepsilon' = \eta' = ad$,
reliqua ut supra (40). (Ex aequat. 20)

45) $fg = a\zeta\eta$, sc. $af^2 = a\zeta\eta$; hinc $\zeta = \eta = f$.

§. 6. Collectis jam sparsis illis coëfficientium valoribus §. praec. erutis, colligitur

1) $\alpha = \beta' = \alpha'' = \beta'' = \gamma'' = \delta''' = \alpha^{IV} = \delta^{IV} = \varepsilon^{IV} = \beta^{V} = \delta^{V} = \zeta^{V}$
$= \alpha^{VI} = \beta^{VI} = \gamma^{VI} = \delta^{VI} = \varepsilon^{VI} = \zeta^{VI} = \eta^{VI} = 1$,
partim ex hypothesibus, partim ex aeq. 35, 37; partim denique ex aequatione $\alpha = \beta' = \gamma'' = \delta'' \ldots = 1$.

2) $\alpha' = \gamma' = \alpha''' = \varepsilon''' = \alpha^{V} = \eta^{V} = \gamma^{V} = \varepsilon^{V} = a$ (28, 38)

3) $\beta = \gamma = \beta''' = \zeta''' = \beta^{IV} = \eta^{IV} = \gamma^{IV} = \zeta^{IV} = b$ (30, 34, 36, 39)

4) $\delta = \varepsilon = \delta' = \zeta' = \delta'' = \eta'' = \varepsilon'' = \zeta'' = d$ (32, 41, 42, 43)

Digitized by Google

5) $\gamma''' = \eta''' = ab$ (40) $= c$
6) $\varepsilon' = \eta' = ad$ (44) $= e$
7) $\zeta = \eta = bd$ (45) $= f$.

Patet ratio, cur (c §. 5. 40, 44) e pluribus solutionibus hic (N°. 5, 6) $\gamma''' = \eta'''$, $\varepsilon' = \eta'$, posuerim; reliquae nempe in radicem producta ab^2, a^2b, ad^2, a^2d introduxissent, unde in ipso quadrato exorti fuissent factores a^2b^4, a^4b^2, a^2d^4, a^4d^2, cum coëfficiente eiusdem quadrati combinandi. Eadem ex causa ex aequatione $a^2 = \gamma^{\nu}\varepsilon^{\nu}$ concludimus $\gamma^{\nu} = \varepsilon^{\nu} = a$ potius quam $\gamma^{\nu} = [\frac{1}{a^2}]$, $\varepsilon^{\nu} = [\frac{a^2}{1}]$ ne alios factores in radicum expressiones introducamus, quam quos *lineares* dicunt.

§. 7. Cum c per ab expressum e calculo eliminare liceat, occupet d illius locum, i. e. pro d scribatur ubique e, adeoque pro e, ac; pro f, bc atque pro $g = af$ (§. 5. 24) scribatur abc; tum habebimus, rite factis substitutionibus et mutatis mutandis:

$(P^2 + aQ^2 + bR^2 + abS^2 + cT^2 + acU^2 + bcV^2 + abcX^2)$ mult. per
$(p^2 + aq^2 + br^2 + abs^2 + ct^2 + acu^2 + bcv^2 + abcx^2) =$
$(Pp + aQq + bRr + abSs + cTt + acUu + bcVv + abcXx)^2$
$+ a\ (Pq - Qp + bRs - bSr + cTu - cUt + bcVx - bcXv)^2$
$+ b\ (Pr - aQs - Rp + aSq + cTv + acUx + cVt + acXu)^2$
$+ ab\ (Ps + Qr - Rq - Sp + cTx + cUv + cVu + cXt)^2$
$+ c\ (Pt - aQu + bRv + abSx - Tp + aUq + bVr + abXs)^2$
$+ ac\ (Pu + Qt + bRx + bSv - Tq - Up + bVs + bXr)^2$
$+ bc\ (Pv - aQx + Rt + aSu + Tr + aUs - Vp + aXq)^2$
$+ abc\ (Px + Qv + Ru + St + Ts + Ur - Vq - Xp)^2$

e qua formula, proposito nobis problemati satisfaciente, innumera quasi corollaria deduci posse nemo non videt. Dubitarit forsan aliquis num 21 aequationes illae a nobis adhibitae proposito sufficiant, cum in determinandis valoribus α, $\alpha' \ldots \beta$, $\beta' \ldots$ etc. eiusmodi productorum, qualia sunt $QRtx$, $RUvq$, etc. nullam ha-

Digitized by Google

buerimus rationem; dubium vero prorsus evanescet consideranti ope illarum 21 aequationum coëfficientes

v. c. producti Qt, Qx, Rt, Rx, Rv, Rq, Uv, Uq etc. sive

$$\alpha^{IV},\ \alpha^{V},\ \beta^{V},\ \beta^{IV},\ \beta'''',\ \beta'',\ \varepsilon'',\ \varepsilon''''\ \text{etc.}$$

ita esse determinatos, ut cum indole radicis primi quadrati sc. $Pp + a\,Qq + b\,Rr$ etc. conveniant, et ea ex causa necessario inter se convenire, vi antiqui axiomatis: *Quae in eodem tertio etc.* Sic cum sit $QRtx = Qt \,.\, Rx = Qx \,.\, Rt$, oportet sit $e\alpha^{IV}.\beta^{IV} + f\alpha^{V}.\beta^{V} = 0$ sive, mutata notatione $ac\,.\,(+b) + bc\,.\,(+a) = 0$. Item, ob $RUvq = Rv \,.\, Uq = Rq \,.\, Uv$, oportet sit $d\,\beta''''\varepsilon'''' + c\,\beta''\varepsilon'' = 0$, sive mutato charactere, $c\,(+b)\,a + ab\,(+c) = 0$. Huius conclusionis veritatem cum formula inventa ubivis affirmet, pro comperto habemus, eam inter demonstrata locum mereri.

§. 8. Ad faciliorem sequentium intellectum juvabit *schema vacuum* (§. 3.) a *schemate completo* (§. 7.) distinxisse. In utroque autem quodlibet octenorum quadratorum ope initialis membri hunc in modum $\overline{\overline{Pv}}^{2}$, $\overline{\overline{Px}}^{2}$, etc. indicabo. Deinde etiam, vitandae confusionis ergo conveniet tam membris initialibus, Pp, Pq ... quam omnibus quadratorum $\overline{\overline{Pp}}^{2}$ et $\overline{\overline{Pq}}^{2}$ membris signa simplicia tribuere. Proinde etiam membris diagonalibus, factorem p continentibus, eadem competent. Denique pro litteris Q, R, S ... q, r, s ... signa P', P'', P''', P^{IV} ... p', p'', p''', p^{IV} ... introducamus.

His praemissis videamus, quid analogia pro schematis vacui constructione praecipiat.

I. Manifesto erit, si ad sedecim quadrata progrediendum sit,

$$\overline{\overline{Pp}} = Pp + P'p' + P''p'' + + + P^{XV}p^{XV}$$

et $$\overline{\overline{Pp'}} = Pp' - P'p + P''p''' - P'''p'' .. + \div + ... P^{XIV}p^{XV} - P^{XV}p^{XIV}.$$

II. In radice $\overline{\overline{Pp''}}$ necessario etiam occurret membrum $-P''p$; et quoniam $\overline{\overline{Pp'}}$ membra $+P''.p'''$, $-P'p$, continet, quorum pro-

Digitized by Google

ductum est $-P'P''pp'''$, hoc per $-P''p$ divisum praebet quotum $+P'p'''$, opposito cum signo in schemate collocandum, una cum *membro reciproco* $+P'''p'$, quod ubique observandum est. Habetur ita *schema* sedecim membrorum *partiale*,

$$(A') \quad \left\{ \begin{array}{llll} P\,p & + P'\,p' & + P''\,p'' & + P'''\,p''' \\ P\,p' & - P'\,p & + P''\,p''' & - P'''\,p'' \\ P\,p'' & - P'\,p''' & - P''\,p & + P'''\,p' \\ P\,p''' & + P'\,p'' & - P''\,p' & - P'''\,p \end{array} \right\}$$

Nota. Hic et donec contrarium moneatur, suaserim, a *signis* $+$ et $-$, ceu minus essentialibus et quasi schematum replendorum gratia tantisper additis — abstrahatur, cum *membra*, sc. producta formae $P^{(m)}p^{(n)}$, hic imprimis spectanda veniant.

facilis scil. est seriei quartae formatio, si observes, a quolibet $-P^{(n)}p$ sursum et sinistrorsum progredienti indices factorum $p^{(m)}$ eodem ordine occurrere, v. c. a $(-P'''p)$ orienti utraque via inita obviam fiunt factores p, p', p'', p'''; a $(-P''p)$, factores p, p''', p'' et sic porro.

III. Schemati (A) dextram versus jungantur schemata similia, indicibus tantum diversa

$$(B') \quad \left\{ \begin{array}{llll} + P^{IV}\,p^{IV} & + P^{V}\,p^{V} & + P^{VI}\,p^{VI} & + P^{VII}\,p^{VII} \\ + P^{IV}\,p^{V} & - P^{V}\,p^{IV} & + P^{VI}\,p^{VII} & - P^{VII}\,p^{VI} \end{array} \right\} (\pm)$$

$$\text{et } (\mp) \quad \left\{ \begin{array}{llll} + P^{IV}\,p^{VI} & - P^{V}\,p^{VII} & - P^{VI}\,p^{IV} & + P^{VII}\,p^{V} \\ + P^{IV}\,p^{VII} & + P^{V}\,p^{VI} & - P^{VI}\,p^{V} & - P^{VII}\,p^{IV} \end{array} \right\}$$

et duo alia, brevitatis gratia ita delineanda:

$$(C') \quad \left\{ \begin{array}{ll} P^{VIII}\,p^{VIII} & \dots\ P^{XI}\,p^{XI} \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ P^{VIII}\,p^{XI} & \dots\ P^{XI}\,p^{VIII} \end{array} \right\} \text{ et } (D') \quad \left\{ \begin{array}{ll} P^{XII}\,p^{XII} & \dots\ P^{XV}\,p^{XV} \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ P^{XII}\,p^{XV} & \dots\ P^{XV}\,p^{XII} \end{array} \right\}$$

quae quatuor schemata quasi zonam horizontalem primam constituunt.

Digitized by Google

IV. Schemati (A′) subjiciatur (A″) ipsi (B′) diagonaliter oppositum, et in quo factores $p^{(n)}$ eodem ordine, quo in (B′), sese excipiunt, prout factores $P^{(n)}$ ordinem in (A′) observatum sequuntur, signa vero membrum 1^{mum} et 2^{dum}, 3^{tium} et 4^{tum}, etc. jungentia diversa sunt. Erit itaque huius indolis:

$$(A'') \quad \left\{ \begin{array}{llll} P\,p^{IV} & - P'\,p^{V} & \mp P''\,p^{VI} & \mp P'''\,p^{VII} \\ P\,p^{V} & + P'\,p^{IV} & \mp P''\,p^{VII} & \mp P'''\,p^{VI} \\ P\,p^{VI} & - P'\,p^{VII} & \mp p''\,p^{IV} & \pm P'''\,p^{V} \\ P\,p^{VII} & + P'\,p^{VI} & \pm P''\,p^{V} & \pm P'''\,p^{IV} \end{array} \right\}$$

cui, vi naturae membrorum reciprocorum, respondebit schema sequens ad dextram situm:

$$(B'') \quad \left\{ \begin{array}{llll} - P^{IV}\,p & + P^{V}\,p' & \mp P^{VI}\,p'' & \pm P^{VII}\,p''' \\ - P^{IV}\,p' & - P^{V}\,p & \pm P^{VI}\,p''' & \pm P^{VII}\,p'' \\ \pm P^{IV}\,p'' & \mp P^{V}\,p''' & - P^{VI}\,p & + P^{VII}\,p' \\ \mp P^{IV}\,p''' & \mp P^{V}\,p'' & - P^{VI}\,p' & - P^{VII}\,p \end{array} \right\}$$

quod, aeque ac (A″) respectu ipsius B′, exhibet factores $p^{(n)}$ eodem ordine progredientes, quo in diagonaliter opposito A′ conspiciuntur dispositi.

V. Formatis schematis $\left\{ \begin{matrix} A', & B' \\ A'', & B'' \end{matrix} \right\}$, ex eisdem combinatis componitur schema partiale majus, 64 producta continens, quod unica littera $\mathfrak{A}'$ denotabimus. Pari modo erit nobis $\mathfrak{B}'$ idem ac systema $\left\{ \begin{matrix} C', & D' \\ C'', & D'' \end{matrix} \right\}$, $\mathfrak{A}''$ idem ac $\left\{ \begin{matrix} A''', & B''' \\ A^{IV}, & B^{IV} \end{matrix} \right\}$ et $\mathfrak{B}''$ idem ac $\left\{ \begin{matrix} C''', & D''' \\ C^{IV}, & D^{IV} \end{matrix} \right\}$ quorum quidem systematum $\mathfrak{B}'$ ab $\mathfrak{A}'$, $\mathfrak{B}''$ ab $\mathfrak{A}''$ indicum tantum valore differre censendum est, abstractione a signis posthac seorsim determinandis, facta. Continent autem systemata $\mathfrak{A}'$, $\mathfrak{B}'$, $\mathfrak{A}''$, $\mathfrak{B}''$ octo series horizontales, quarum quaelibet ipsa est octimembris; quare coëuntibus quaternis hisce systematis prodit systema sedecim serierum, quarum quaelibet sedecim est membrorum.

VI. Cum autem indices factorum $P^{(n)}$ ordine naturali progre-

Digitized by Google

diantur, his factoribus facile supplendis (et etiam tamdiu signis) omissis, sequens nanciscimur schema 16 serierum:

Series	1.	a.	b.	ab.	c.	ac.	bc.	abc.	d.	ad.	bd.	abd.	cd.	acd.	bcd.	abcd.	
1	$p_{\cdot}$	p^{I}	p^{II}	p^{III}	p^{IV}	p^{V}	p^{VI}	p^{VII}	p^{VIII}	p^{IX}	p^{X}	p^{XI}	p^{XII}	p^{XIII}	p^{XIV}	p^{XV}	1
2	p^{I}	$p_{\cdot}$	p^{III}	p^{II}	p^{V}	p^{IV}	p^{VII}	p^{VI}	p^{IX}	p^{VIII}	p^{XI}	p^{X}	p^{XIII}	p^{XII}	p^{XV}	p^{XIV}	a
3	p^{II}	p^{III}	$p_{\cdot}$	p^{I}	p^{VI}	p^{VII}	p^{IV}	p^{V}	p^{X}	p^{XI}	p^{VIII}	p^{IX}	p^{XIV}	p^{XV}	p^{XII}	p^{XIII}	b
4	p^{III}	p^{II}	p^{I}	$p_{\cdot}$	p^{VII}	p^{VI}	p^{V}	p^{IV}	p^{XI}	p^{X}	p^{IX}	p^{VIII}	p^{XV}	p^{XIV}	p^{XIII}	p^{XII}	ab
5	p^{IV}	p^{V}	p^{VI}	p^{VII}	$p_{\cdot}$	p^{I}	p^{II}	p^{III}	p^{XII}	p^{XIII}	p^{XIV}	p^{XV}	p^{VIII}	p^{IX}	p^{X}	p^{XI}	c
6	p^{V}	p^{IV}	p^{VII}	p^{VI}	p^{I}	$p_{\cdot}$	p^{III}	p^{II}	p^{XIII}	p^{XII}	p^{XV}	p^{XIV}	p^{IX}	p^{VIII}	p^{XI}	p^{X}	ac
7	p^{VI}	p^{VII}	p^{IV}	p^{V}	p^{II}	p^{III}	$p_{\cdot}$	p^{I}	p^{XIV}	p^{XV}	p^{XII}	p^{XIII}	p^{X}	p^{XI}	p^{VIII}	p^{IX}	bc
8	p^{VII}	p^{VI}	p^{V}	p^{IV}	p^{III}	p^{II}	p^{I}	$p_{\cdot}$	p^{XV}	p^{XIV}	p^{XIII}	p^{XII}	p^{XI}	p^{X}	p^{IX}	p^{VIII}	abc
9	p^{VIII}	p^{IX}	p^{X}	p^{XI}	p^{XII}	p^{XIII}	p^{XIV}	p^{XV}	$p_{\cdot}$	p^{I}	p^{II}	p^{III}	p^{IV}	p^{V}	p^{VI}	p^{VII}	d
10	p^{IX}	p^{VIII}	p^{XI}	p^{X}	p^{XIII}	p^{XII}	p^{XV}	p^{XIV}	p^{I}	$p_{\cdot}$	p^{III}	p^{II}	p^{V}	p^{IV}	p^{VII}	p^{VI}	ad
11	p^{X}	p^{XI}	p^{VIII}	p^{IX}	p^{XIV}	p^{XV}	p^{XII}	p^{XIII}	p^{II}	p^{III}	$p_{\cdot}$	p^{I}	p^{VI}	p^{VII}	p^{IV}	p^{V}	bd
12	p^{XI}	p^{X}	p^{IX}	p^{VIII}	p^{XV}	p^{XIV}	p^{XIII}	p^{XII}	p^{III}	p^{II}	p^{I}	$p_{\cdot}$	p^{VII}	p^{VI}	p^{V}	p^{IV}	abd
13	p^{XII}	p^{XIII}	p^{XIV}	p^{XV}	p^{VIII}	p^{IX}	p^{X}	p^{XI}	p^{IV}	p^{V}	p^{VI}	p^{VII}	$p_{\cdot}$	p^{I}	p^{II}	p^{III}	cd
14	p^{XIII}	p^{XII}	p^{XV}	p^{XIV}	p^{IX}	p^{VIII}	p^{XI}	p^{X}	p^{V}	p^{IV}	p^{VII}	p^{VI}	p^{I}	$p_{\cdot}$	p^{III}	p^{II}	acd
15	p^{XIV}	p^{XV}	p^{XII}	p^{XIII}	p^{X}	p^{XI}	p^{VIII}	p^{IX}	p^{VI}	p^{VII}	p^{IV}	p^{V}	p^{II}	p^{III}	$p_{\cdot}$	p^{I}	bcd
16	p^{XV}	p^{XIV}	p^{XIII}	p^{XII}	p^{XI}	p^{X}	p^{IX}	p^{VIII}	p^{VII}	p^{VI}	p^{V}	p^{IV}	p^{III}	p^{II}	p^{I}	$p_{\cdot}$	abcd
	P,	P^{I},	P^{II},	P^{III},	P^{IV},	P^{V},	P^{VI},	P^{VII},	P^{VIII},	P^{IX},	P^{X},	P^{XI},	P^{XII},	P^{XIII},	P^{XIV},	P^{XV},	

cujus e conspectu optime et facillime methodum ulterius progrediendi addiscere possumus. Series 1, 4; 2, 3; 5, 8; 6, 7; 9, 12; 10, 11; 13, 16; 14, 15; monstrant in quolibet quadrato factorum $p^{(n)}$ inversum ordinem. Subjunximus simul factores $P^{(m)}$ factoribus $p^{(n)}$ in eadem serie verticali positis jungendos, nec non a latere ordinem coëfficientium 1, *a*, *b*, *ab*, ... cujuslibet $Pp^{\overline{(n)}^s}$ exhibuimus.

VII. Quaeritur jam *factor* producti $P^{(m)}\,p^{(n)}$, v. c. $P^{XI}\,p^{XIII}$, quem eundem esse novimus cum factore producti $P^{(n)}\,p^{(m)}$, hic $P^{XIII}\,p^{XI}$, opposito tamen signo affecti. Reperiuntur autem haec producta in serici horizontali 7ma, cujus coëfficiens est *bc*. Si vero e serie prima, cui superscriptos videmus eosdem subscriptis sibi terminis seriei horizontalis 1^{ae}. congruos coëfficientes, membrorum $P^{XI}\,p^{XI}$ et $P^{XIII}\,p^{XIII}$ factores *abd* et *acd* excerpamus, ex horum combinatione nanciscimur productum a^2bcd^2, quod per coëfficientem (*bc*) seriei 7^{mae} divisum praebet quotum in binos factores (ex causis supra allatis invicem aequales, adeoque $= ad$) dispescendum. Erit igitur

Digitized by Google

membrorum radix $\overline{Pp^{VI}}$ alterum $\pm ad\, P^{XI}p^{XIII}$, alterum $\mp ad\, P^{XIII}p^{XI}$. His ubique observatis determinatio coëfficientium, quibus radicum singuli termini afficiendi sint, nulla plane difficultate laborat.

VIII. *Signa* quod attinet, notetur, *horum pleraque esse ambigua*. Simplicitatis observandae gratia statuimus omnia primae seriei tam horizontalis, quam verticalis, signa esse positiva; secundae seriei horizontalis adeo alternantia sunt necessario. Deinde vero singula diagonalis Pp — $P^{XV}p$ membra, formae $P^{(m)}p$, sunt negativa; unde sequitur, a diagonali quocunque membro $P^{(m)}p$ $\left[\begin{smallmatrix}\text{sinistrorsum}\\ \text{dextrorsum}\end{smallmatrix}\right]$ et $\left[\begin{smallmatrix}\text{sursum}\\ \text{deorsum}\end{smallmatrix}\right]$ progredientibus occurrere membra formae $P^{(m+z)}p^{(n)}$ et $P^{(m)}p^{(a)}$, z terminis a diagonali illo distantia et oppositis signis affecta. Reliqua omnia manent ambigua, ita tamen ut signi dato cuidam membro respondentis electio aliorum quorundam membrorum signa determinet. Quod, quo clarius monstremus, ponamus characteribus α, β, γ, ... denotari signa quaelibet $+$ aut $-$, his vero $\overline{\alpha}$, $\overline{\beta}$, $\overline{\gamma}$, ... eorundem signorum opposita, ita ut sit $\overline{\alpha}$ idem quod $-\alpha$; $-\overline{\beta}$ i. qu. β; etc:

Jam, cum praeter diagonalem et series tam horizontales quam verticales 1^{mam} et 2^{dam}, reliquarum signa nondum determinata esse fingamus, proposita signi membro $P^{IV}p^{VI}$ (ser. hor. 3) respondentis determinatione, sit idem α. Occurrit autem in eadem serie necessario membrum reciprocum $P^{VI}p^{IV}$, cujus ideo signum erit $-\alpha$ sive $\overline{\alpha}$. Cum vero horum membrorum productum $P^{IV}p^{VI}.\ P^{VI}p^{IV} = P^{IV}p^{IV}.\ P^{VI}p^{VI}$ etiam in seriei prima horizontali occurrat et vi assumtionis sit positivum, perspicitur $\tau\grave{o} - \alpha^2$ aequivalere $\tau\tilde{\omega} -$, adeoque $\tau\grave{o}\ \alpha^2\ \tau\tilde{\omega} +$, ideoque α significare signum ambiguum $\pm$. Nunc vero etiam in serie verticali 3^{tia} occurrunt membra $P''p^{VI}$ et $P'''p^{IV}$, quibus, ob ante notata, tribuenda sunt signa $\overline{\alpha}$ et α (serierum hor. 5 et 7). In iisdem (5 et 7) iterum spectanda sunt membra reciproca $P^{VI}p''$ et $P^{IV}p''$, signis contrariis α et $\overline{\alpha}$ afficienda. Hinc perspicitur, introducendo membro $\pm P^{IV}p^{VI}$, obtineri

Digitized by Google

alia plura, scil. $\mp P^{VI} p^{IV}$, $\mp P'' p^{VI}$, $\pm P^{VI} p''$, $P'' p^{IV}$, $P^{IV} p''$. Igitur nec signorum determinatio, prolixitatem hic prout in factorum determinatione si exceperis, nulli prorsus defficultati est obnoxia.

§. 9. Ita viam stratam patefecimus ad demonstrationem insignis hujus theorematis:

Si fuerit $P^2 + a P'^2 + b P''^2 + a b P'''^2 + c P^{IV\,2} + a c P^{V2} + b c P^{VI^2} + a b c P^{VII^2} + d P^{VIII^2} + + \ldots$ *expressio* 2^n *terminorum, atque* $p^2 + a p'^2 + b p''^2 + + \ldots$ *alia expressio ipsi* $P^2 + a P'^2 + b P''^2 + + \ldots$ *perfecte similis, erit etiam productum harum expressionum ipsis factoribus prorsus simile.*

Quare, si forte hujus similitudinis efficiendae methodum in antecedentibus adumbrasse successerit, nexum theorematum a viris illis immortalibus, ill. *Eulero* et ill. *Lagrangio* prolatorum cum veritate universali ostendisse contenti, finem huic tractatiunculae ponimus, nobis congratulaturi, si levi hoc tentamine scientiae analyticae amatores ad exemplaria summorum horum magistrorum exemplaria nocturna, diurna manu versanda excitaverimus.

Digitized by Google